

School of Computer Science and Artificial Intelligence

WEEK-9

Assignment #18

Program : B. Tech (CSE)
Specialization :AIML
Course Title : Competitive Programming
Course Code : 23CS302PC305
Semester : VI
Academic Session : 2025-2026
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Batch No. : 33

Assignment 3: Pattern Prefix Matching in Log Analysis (Z-Algorithm)

A system administrator analyzes system logs to detect whether a specific error pattern appears as a prefix at any position in the log file.

Task

- Input

Pattern: abcaby Output

PYTHON CODE:

```
main.py [ ] [ ] Share Run Output  
1 def build_z_array(s):  
2     n = len(s)  
3     z = [0] * n  
4     l, r = 0, 0  
5  
6     for i in range(1, n):  
7         if i <= r:  
8             z[i] = min(r - i + 1, z[i - 1])  
9  
10            while i + z[i] < n and s[z[i]] == s[i + z[i]]:  
11                z[i] += 1  
12  
13            if i + z[i] - 1 > r:  
14                l, r = i, i + z[i] - 1  
15  
16        return z  
17  
18  
19 def find_pattern(text, pattern):  
20     combined = pattern + "$" + text  
21     z = build_z_array(combined)  
22     m = len(pattern)  
23  
24     result = []
```

```
main.py  [Icons]  Share  Run  Output

21  z = build_z_array(combined)
22  m = len(pattern)
23
24  result = []
25
26  for i in range(len(z)):
27      if z[i] == m:
28          index = i - m + 1
29          result.append(index)
30
31  return result
32
33
34  # ---- MAIN ----
35  text = input("Log text: ").strip()
36  pattern = input("Pattern: ").strip()
37
38  matches = find_pattern(text, pattern)
39
40  if matches:
41      for idx in matches:
42          print("Pattern found at index", idx)
43  else:
44      print("No match found")
```

Log text: abxabcabcaby
Pattern: abcaby
Pattern found at index 6
==> Code execution Successful

JAVA CODE:

```
ZAlgorithmSearch.java
1- import java.util.*;
2-
3- public class ZAlgorithmSearch {
4-
5-     static int[] buildZArray(String s) {
6-         int n = s.length();
7-         int[] z = new int[n];
8-         int l = 0, r = 0;
9-
10-        for (int i = 1; i < n; i++) {
11-            if (i <= r) {
12-                z[i] = Math.min(r - i + 1, z[i - 1]);
13-            }
14-
15-            while (i + z[i] < n && s.charAt(z[i]) == s.charAt(i + z[i])) {
16-                z[i]++;
17-            }
18-
19-            if (i + z[i] - 1 > r) {
20-                l = i;
21-                r = i + z[i] - 1;
22-            }
23-        }
24-    }
}
```

Log text: abxabcabcaby
Pattern: abcaby
Pattern found at index 6
=== Code Execution Successful ===

```
ZAlgorithmSearch.java
43- }
44-
45- public static void main(String[] args) {
46-     Scanner sc = new Scanner(System.in);
47-
48-     System.out.print("Log text: ");
49-     String text = sc.next();
50-
51-     System.out.print("Pattern: ");
52-     String pattern = sc.next();
53-
54-     List<Integer> matches = findPattern(text, pattern);
55-
56-     if (!matches.isEmpty()) {
57-         for (int idx : matches) {
58-             System.out.println("Pattern found at index " + idx);
59-         }
60-     } else {
61-         System.out.println("No match found");
62-     }
63-
64-     sc.close();
65- }
66- }
```

Log text: abxabcabcaby
Pattern: abcaby
Pattern found at index 6
=== Code Execution Successful ===

Search

